

SHAMIK BASU

Data Scientist

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PROFESSIONAL SUMMARY

Data scientist with 3+ years shipping production machine learning pipelines, LLM systems, and analytics workflows with measurable business impact. Cut operational costs by 72% on production RAG processing 5M+ records/month and improved fraud identification accuracy by 27% while reducing false positives. Skilled in Python and SQL, statistics, experimentation, causal inference, econometrics, machine learning, and forecasting.

RELEVANT SKILLS

Languages & Databases: Python, SQL, R, C/C++, PostgreSQL, MongoDB, Spark, Trino, Apache Kafka

ML & Analytics: Machine learning, statistics, econometrics, causal inference, A/B experimentation, feature engineering, logistic regression, NLP, NER, forecasting, quantitative analysis, Tableau, Power BI

Deep Learning & LLMs: PyTorch, TensorFlow, Hugging Face Transformers, LoRA fine-tuning, RAG, LangChain, multi-agent orchestration

Cloud: AWS (S3, Lambda, DynamoDB), Docker, Kubernetes, REST APIs, CI/CD, Azure Synapse, Langfuse observability, Git

Leadership: VP of GRIDS, 250-member student-led graduate data science club at USC, | Graduate Mentor for USC Viterbi Engineering

WORK EXPERIENCE

Data Science Associate Intern (Part-Time) | Liquidity AI Capital | Los Angeles, CA

April 2026 - May 2026

- Deployed production SLM inference in Docker by fine-tuning open-source models end-to-end (tokenization and LoRA adapter training) with optimized domain-specific inference on financial data.

Data Science Associate Intern (Part-Time) | KCC Capital Partners | Los Angeles, CA

January 2026 - March 2026

- Automated M&A target sourcing across active mandates by building an agentic LangChain pipeline (Python) chaining LLM entity classification, web scraping, and structured extraction to identify and rank strategic buyers.

Data Scientist | Bajaj Finserv Health | Pune, India

November 2023 - December 2024

- Cut operational costs by 72% at 92% extraction accuracy by architecting a production RAG pipeline (LangChain + GPT-3.5 Turbo + REST APIs) processing 5M+ medical records/month, replacing a manual clinical workflow.
- Reduced GPU compute waste by 15% by building Langfuse observability for agentic pipelines tracking token utilization, output drift, and latency to prioritize production optimizations.
- Achieved zero-downtime real-time inference by migrating batch cronjobs to an Apache Kafka event-driven architecture, shipping in two-week agile sprints with automated integration tests.

Associate Data Scientist | Bajaj Finserv Health | Pune, India

July 2022 - October 2023

- Boosted fraud identification accuracy by 27% while reducing false positives by redesigning the production NER pipeline, retraining entity extraction models, and improving name-matching heuristics.
- Improved business performance and operational efficiency by 22% by building a supervised machine learning model (logistic regression + feature engineering), statistically validating quantitative research findings, and operationalizing workforce predictions.

Data Engineer Intern | Bajaj Finserv Health | Pune, India

January 2022 - June 2022

- Scaled ETL and analytics to 10M+ records across 200+ features by designing a distributed system (C++, Trino, Docker) with optimized SQL, analytical query processing, and disaster recovery on an enterprise SQL cluster.

PROJECTS

Custom CUDA Library for CNN Pre-Processing

January 2026 - February 2026

- Achieved 115x CPU speedup at N=1024 (Tesla T4) and 14x PyTorch data preprocessing speedup by profiling and optimizing shared-memory tiled CUDA kernels (C/C++) for matrix multiplication and image convolution.
- Delivered Python-bindable .so libraries for GPU-accelerated data preprocessing, integrating optimized kernels into ML pipelines without rewriting downstream Python code.

Agentic Trend Orchestrator

May 2026 - June 2026

- Owned full SDLC through CI/CD deployment by building a multi-agent AI system (FastAPI + Next.js + Docker) with persistent memory and LLM-driven task orchestration.
- Improved agent reliability by iterating on task completion rate, tool-use accuracy, and semantic drift metrics across orchestration workflows.

EcoMateAI

May 2025 - June 2025

- Won 1st place at UCLA SAIRS Hackathon (Microsoft, IBM, NVIDIA, Google, LinkedIn) by shipping a multimodal agentic AI platform (Gemini 2.5 Flash) in 24 hours.
- Ingested text, images, and structured data through multimodal data preprocessing and agent workflows to generate personalized sustainability recommendations.

EDUCATION

University of Southern California | Masters, Applied Data Science

January 2025 - December 2026

Coursework: Deep Learning, Machine Learning, Data Mining (MapReduce), Data Management

SRM Institute of Science & Technology | Bachelor, Computer Science

May 2018 - June 2022

Coursework: Algorithms, Artificial Intelligence, Computer Vision, Probability & Queueing Theory